

Recursive Sequences

1. The sequence $a_1, a_2, a_3, \dots$ satisfies $a_n = a_{n-1} + 2n - 1$ with $a_1 = 1$. Find a_{100} .
2. A sequence is defined by $a_1 = 1$, $a_2 = 1$, and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$ (the Fibonacci sequence). Find the last digit of a_{100} .
3. Define $a_1 = 1$ and $a_n = 7^{a_{n-1}}$ for $n \geq 2$. Find the last two digits of a_{1000} .
4. Let $a_1 = 7$ and $a_n = a_{n-1}^2 - 2$ for $n \geq 2$. Find the last digit of a_{50} .
5. Define $a_1 = 1$ and $a_{n+1} = \frac{a_n + 1}{a_n + 2}$ for $n \geq 1$. Find a_{2025} .
6. Let $a_1 = 1$, $a_2 = 2$, $a_3 = 3$, and $a_n = a_{n-1} - a_{n-2} + a_{n-3}$ for $n \geq 4$. Find a_{2024} .
7. Define $a_1 = 2$ and $a_{n+1} = \frac{2a_n}{a_n + 2}$ for $n \geq 1$. Find a closed form for a_n and compute a_{100} .
8. Define $a_0 = 0$, $a_1 = 1$, and $a_{n+1} = 3a_n - a_{n-1}$ for $n \geq 1$. Prove that $a_n^2 + a_{n+1}^2 - a_n a_{n+1}$ is constant, and find the value.
9. Let $b_n = a_{n+1}/a_n$ where $a_1 = 1$, $a_2 = 3$, and $a_{n+1} = 4a_n - a_{n-1}$. As $n \rightarrow \infty$, b_n approaches a limit L . Find L .
10. Define $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$ for $n \geq 1$. Find $\lim_{n \rightarrow \infty} a_n$.
11. Let $a_1 = 1$, $a_2 = 5$, and $a_{n+2} = 5a_{n+1} - a_n$ for $n \geq 1$. Prove that a_n is always an odd integer, and find $a_{10} \pmod{7}$.
12. Let $a_1 = 2$, $a_2 = 2$, and $a_n = a_{n-1} \cdot a_{n-2}$ for $n \geq 3$. Find the largest power of 2 dividing a_{20} .
13. Define $a_1 = 1$, $a_2 = 0$, and $a_{n+2} = 2a_{n+1} - a_n + 2$ for $n \geq 1$. For how many integers $1 \leq n \leq 100$ is a_n divisible by 4?
14. Let $a_0 = 3$, $a_1 = 3$, and $a_n = a_{n-1}^2 - 2$ for $n \geq 2$. Find $a_{100} \pmod{9}$.
15. A sequence satisfies $a_1 = 1$ and

$$a_{n+1} = \frac{a_n}{1 + n a_n}$$

for $n \geq 1$. Find a_{100} .

16. Define $a_1 = 19$, $a_2 = 98$, and $a_{n+2} = a_n - \frac{1}{a_{n+1}}$ for $n \geq 1$. Show that $a_n > 0$ for all n , and find $\lfloor a_{1999} \rfloor$. (Assume without proof that the sequence decreases slowly.)

17. Let $a_1 = a_2 = 1$ and $a_{n+2} = a_{n+1} + a_n + n$ for $n \geq 1$. Find a closed form for a_n .

18. A sequence is defined by $a_1 = \cos \theta$ and

$$a_{n+1} = 2a_n^2 - 1$$

for some fixed $\theta \in (0, \pi)$. Find a_n in terms of θ , and determine all θ for which the sequence is periodic with period exactly 3.

19. Let $a_1, a_2, \dots$ be a sequence of positive reals with $a_1 = 1$, $a_2 = 2$, and

$$a_{n+2} = \frac{a_{n+1}^2 + 1}{a_n}$$

for $n \geq 1$. Prove that every term is a positive integer, and find the largest $n \leq 100$ for which $a_n < 1000$.